CORE PURE / MISCELLANEOUS PURE

Method of Differences

Worked solutions 12 questions · 90 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. **(a)** Show that, for $r > 0$,

$$\ln\left(1+\frac{2}{r(r+3)}\right) = \ln\left(\frac{r+1}{r}\right) - \ln\left(\frac{r+3}{r+2}\right) \quad \textbf{[2]}$$

(b) Hence, using the method of differences, show that for integer $n \geq 1$,

$$\sum_{r=1}^{n} \ln\left(1+\frac{2}{r(r+3)}\right) = \ln\left(\frac{3(n+1)}{n+3}\right) \quad \textbf{[4]}$$

Solution

(a) First simplify the term inside the logarithm:

$$1+\frac{2}{r(r+3)} = \frac{r(r+3)+2}{r(r+3)} = \frac{r^2+3r+2}{r(r+3)} = \frac{(r+1)(r+2)}{r(r+3)} = \frac{r+1}{r}\times\frac{r+2}{r+3}$$

So

$$\begin{aligned}\ln\left(1+\frac{2}{r(r+3)}\right) &= \ln\left(\frac{r+1}{r}\right) + \ln\left(\frac{r+2}{r+3}\right)\\ &= \ln\left(\frac{r+1}{r}\right) - \ln\left(\frac{r+3}{r+2}\right)\end{aligned}$$

as required.

(b) Let

$$S_n = \sum_{r=1}^{n} \ln\left(1+\frac{2}{r(r+3)}\right)$$

Using part (a):

$$\begin{aligned}S_n &= \sum_{r=1}^{n}\left[\ln\left(\frac{r+1}{r}\right) - \ln\left(\frac{r+3}{r+2}\right)\right]\\ &= \left(\ln 2 - \ln\frac{4}{3}\right) + \left(\ln\frac{3}{2} - \ln\frac{5}{4}\right) + \left(\ln\frac{4}{3} - \ln\frac{6}{5}\right) + \cdots + \left(\ln\frac{n+1}{n} - \ln\frac{n+3}{n+2}\right)\end{aligned}$$

Now write the positive and negative terms together:

$$\begin{aligned}S_n = \ln 2 + \ln\frac{3}{2} + \cancel{\ln\frac{4}{3}} + \cancel{\ln\frac{5}{4}} + \cancel{\ln\frac{6}{5}} + \cdots + \cancel{\ln\frac{n+1}{n}}\\ - \cancel{\ln\frac{4}{3}} - \cancel{\ln\frac{5}{4}} - \cancel{\ln\frac{6}{5}} - \cdots - \cancel{\ln\frac{n+1}{n}} - \ln\frac{n+2}{n+1} - \ln\frac{n+3}{n+2}\end{aligned}$$

All the coloured middle terms cancel, so

$$\begin{aligned}S_n &= \ln 2 + \ln\frac{3}{2} - \ln\frac{n+2}{n+1} - \ln\frac{n+3}{n+2}\\ &= \ln\left(2\times\frac{3}{2}\times\frac{n+1}{n+2}\times\frac{n+2}{n+3}\right)\\ &= \ln\left(\frac{3(n+1)}{n+3}\right)\end{aligned}$$

Therefore

$$\sum_{r=1}^{n} \ln\left(1 + \frac{2}{r(r+3)}\right) = \ln\left(\frac{3(n+1)}{n+3}\right)$$

2. **(a)** By expressing $\dfrac{1}{r(r+2)}$ in partial fractions, show that

$$\sum_{r=1}^{n} \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)} \qquad \textbf{[5]}$$

(b) Hence determine the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+2)} \qquad \textbf{[2]}$$

Solution

(a) Let

$$S_n = \sum_{r=1}^{n} \frac{1}{r(r+2)}$$

First express the general term in partial fractions:

$$\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2}$$

So

$$1 = A(r+2) + Br = (A+B)r + 2A$$

Comparing coefficients gives

$$A + B = 0, \qquad 2A = 1$$

Hence

$$A = \frac{1}{2}, \qquad B = -\frac{1}{2}$$

Therefore

$$\frac{1}{r(r+2)} = \frac{1}{2}\left(\frac{1}{r} - \frac{1}{r+2}\right)$$

Substituting into the sum:

$$\begin{aligned} S_n &= \frac{1}{2}\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+2}\right) \\ &= \frac{1}{2}\left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+2}\right)\right] \end{aligned}$$

Now use the method of differences:

$$\begin{aligned} S_n &= \frac{1}{2}\left(1+\frac{1}{2}+\cancel{\frac{1}{3}}+\cancel{\frac{1}{4}}+\cdots+\cancel{\frac{1}{n}}-\cancel{\frac{1}{3}}-\cancel{\frac{1}{4}}-\cdots-\cancel{\frac{1}{n}}-\frac{1}{n+1}-\frac{1}{n+2}\right) \\ &= \frac{1}{2}\left(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2}\right) \\ &= \frac{1}{2}\left(\frac{3}{2}-\frac{1}{n+1}-\frac{1}{n+2}\right) \\ &= \frac{3}{4}-\frac{1}{2(n+1)}-\frac{1}{2(n+2)} \end{aligned}$$

Hence

$$\sum_{r=1}^{n}\frac{1}{r(r+2)} = \frac{3}{4}-\frac{1}{2(n+1)}-\frac{1}{2(n+2)}$$

(b) From part (a):

$$\sum_{r=1}^{n}\frac{1}{r(r+2)} = \frac{3}{4}-\frac{1}{2(n+1)}-\frac{1}{2(n+2)}$$

As $n \to \infty$

$$\frac{1}{2(n+1)} \to 0, \qquad \frac{1}{2(n+2)} \to 0$$

so

$$\sum_{r=1}^{\infty}\frac{1}{r(r+2)} = \frac{3}{4}$$

3. Show that

$$\frac{1}{1\times 4}+\frac{1}{2\times 5}+\frac{1}{3\times 6}+\cdots+\frac{1}{n(n+3)}=\frac{11}{18}-\frac{an^2+bn+c}{3(n+1)(n+2)(n+3)}$$

where a, b and c are constants to be found. **[5]**

Solution

Let

$$S_n=\frac{1}{1\times 4}+\frac{1}{2\times 5}+\frac{1}{3\times 6}+\cdots+\frac{1}{n(n+3)}=\sum_{r=1}^{n}\frac{1}{r(r+3)}$$

Use partial fractions on the general term:

$$\frac{1}{r(r+3)}=\frac{A}{r}+\frac{B}{r+3}$$

So

$$1=A(r+3)+Br=(A+B)r+3A$$

Comparing coefficients:

$$A+B=0,\qquad 3A=1$$

Hence

$$A=\frac{1}{3},\qquad B=-\frac{1}{3}$$

Therefore

$$\frac{1}{r(r+3)}=\frac{1}{3}\left(\frac{1}{r}-\frac{1}{r+3}\right)$$

So

$$S_n=\frac{1}{3}\sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+3}\right)$$

Now write out the terms to use the method of differences:

$$S_n=\frac{1}{3}\Bigg[1+\frac{1}{2}+\frac{1}{3}+\cancel{\frac{1}{4}}+\cancel{\frac{1}{5}}+\cdots+\cancel{\frac{1}{n}}$$

$$-\cancel{\frac{1}{4}}-\cancel{\frac{1}{5}}-\cdots-\cancel{\frac{1}{n}}-\frac{1}{n+1}-\frac{1}{n+2}-\frac{1}{n+3}\Bigg]$$

Everything from $\frac{1}{4}$ to $\frac{1}{n}$ cancels, leaving

$$S_n=\frac{1}{3}\left(1+\frac{1}{2}+\frac{1}{3}-\frac{1}{n+1}-\frac{1}{n+2}-\frac{1}{n+3}\right)$$

Now

$$1+\frac{1}{2}+\frac{1}{3}=\frac{6+3+2}{6}=\frac{11}{6}$$

Hence

$$S_n = \frac{11}{18} - \frac{1}{3}\left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}\right)$$

Combine the three fractions:

$$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} = \frac{(n+2)(n+3) + (n+1)(n+3) + (n+1)(n+2)}{(n+1)(n+2)(n+3)}$$

Expand the numerator:

$$(n+2)(n+3) = n^2 + 5n + 6$$

$$(n+1)(n+3) = n^2 + 4n + 3$$

$$(n+1)(n+2) = n^2 + 3n + 2$$

So

$$(n+2)(n+3) + (n+1)(n+3) + (n+1)(n+2) = 3n^2 + 12n + 11$$

Therefore

$$S_n = \frac{11}{18} - \frac{3n^2 + 12n + 11}{3(n+1)(n+2)(n+3)}$$

Comparing with

$$\frac{11}{18} - \frac{an^2 + bn + c}{3(n+1)(n+2)(n+3)}$$

we get

$$a = 3, \qquad b = 12, \qquad c = 11$$

So the required result is

$$\frac{1}{1\times 4} + \frac{1}{2\times 5} + \frac{1}{3\times 6} + \cdots + \frac{1}{n(n+3)} = \frac{11}{18} - \frac{3n^2 + 12n + 11}{3(n+1)(n+2)(n+3)}$$

4. Use the method of differences to show that

$$\sum_{r=1}^{n} \frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{n(an+b)}{(2n+1)(2n+3)}$$

where a and b are integers to be determined. **[6]**

Solution

Let

$$S_n = \sum_{r=1}^{n} \frac{12}{(2r-1)(2r+1)(2r+3)}$$

Write the general term in partial fractions:

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{A}{2r-1} + \frac{B}{2r+1} + \frac{C}{2r+3}$$

Using the cover-up rule, with $r = \frac{1}{2}$, $r = -\frac{1}{2}$ and $r = -\frac{3}{2}$ in turn:

$$A = \frac{12}{2 \times 4} = \frac{3}{2}, \qquad B = \frac{12}{(-2)(2)} = -3, \qquad C = \frac{12}{(-4)(-2)} = \frac{3}{2}$$

So

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{3}{2(2r-1)} - \frac{3}{2r+1} + \frac{3}{2(2r+3)}$$

It is convenient to rewrite this as

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{3}{2}\left(\frac{1}{2r-1} - \frac{1}{2r+1}\right) - \frac{3}{2}\left(\frac{1}{2r+1} - \frac{1}{2r+3}\right)$$

Hence

$$\begin{aligned} S_n &= \sum_{r=1}^{n} \frac{12}{(2r-1)(2r+1)(2r+3)} \\ &= \frac{3}{2}\sum_{r=1}^{n}\left(\frac{1}{2r-1} - \frac{1}{2r+1}\right) - \frac{3}{2}\sum_{r=1}^{n}\left(\frac{1}{2r+1} - \frac{1}{2r+3}\right) \end{aligned}$$

Expanding both sums:

$$\begin{aligned} S_n = \frac{3}{2}&\left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)\right] \\ &- \frac{3}{2}\left[\left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{9}\right) + \cdots + \left(\frac{1}{2n+1} - \frac{1}{2n+3}\right)\right] \end{aligned}$$

So

$$\begin{aligned} S_n = \frac{3}{2}&\left[1 - \tfrac{1}{3} + \tfrac{1}{3} - \tfrac{1}{5} + \tfrac{1}{5} - \tfrac{1}{7} + \tfrac{1}{7} - \cdots - \frac{1}{2n+1}\right] \\ &- \frac{3}{2}\left[\frac{1}{3} - \tfrac{1}{5} + \tfrac{1}{5} - \tfrac{1}{7} + \tfrac{1}{7} - \cdots - \frac{1}{2n+3}\right] \end{aligned}$$

Hence

$$\begin{aligned} S_n &= \frac{3}{2}\left(1 - \frac{1}{2n+1}\right) - \frac{3}{2}\left(\frac{1}{3} - \frac{1}{2n+3}\right) \\ &= \frac{3}{2} - \frac{3}{2(2n+1)} - \frac{1}{2} + \frac{3}{2(2n+3)} \\ &= 1 + \frac{3}{2}\left(\frac{1}{2n+3} - \frac{1}{2n+1}\right) \\ &= 1 - \frac{3}{(2n+1)(2n+3)} \\ &= \frac{(2n+1)(2n+3) - 3}{(2n+1)(2n+3)} \\ &= \frac{4n^2 + 8n}{(2n+1)(2n+3)} \\ &= \frac{n(4n+8)}{(2n+1)(2n+3)} \end{aligned}$$

Comparing with

$$\frac{n(an+b)}{(2n+1)(2n+3)}$$

gives

$$a = 4, \qquad b = 8$$

So

$$\sum_{r=1}^{n} \frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{n(4n+8)}{(2n+1)(2n+3)}$$

5. **(a)** Use the method of differences to prove that for $n \geq 1$

$$\sum_{r=1}^{n} \ln\left(1 + \frac{2}{r(r+3)}\right) = \ln\left(\frac{3(n+1)}{n+3}\right) \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\sum_{r=25}^{74} \ln\left(\left(1 + \frac{2}{r(r+3)}\right)^{28}\right)$$

Give your answer in the form $a \ln\left(\frac{b}{c}\right)$ where a, b and c are integers to be determined. **[3]**

Solution

(a) Let

$$S_n = \sum_{r=1}^{n} \ln\left(1 + \frac{2}{r(r+3)}\right)$$

First simplify the term inside the logarithm:

$$\begin{aligned} 1 + \frac{2}{r(r+3)} &= \frac{r(r+3)+2}{r(r+3)} \\ &= \frac{r^2+3r+2}{r(r+3)} \\ &= \frac{(r+1)(r+2)}{r(r+3)} \\ &= \frac{r+1}{r} \times \frac{r+2}{r+3} \end{aligned}$$

So

$$\begin{aligned} \ln\left(1 + \frac{2}{r(r+3)}\right) &= \ln\left(\frac{r+1}{r} \times \frac{r+2}{r+3}\right) \\ &= \ln(r+1) - \ln r + \ln(r+2) - \ln(r+3) \end{aligned}$$

Hence

$$S_n = \sum_{r=1}^{n} \big(\ln(r+1) - \ln r\big) + \sum_{r=1}^{n} \big(\ln(r+2) - \ln(r+3)\big)$$

Now expand each sum using the method of differences.

For the first sum:

$$\begin{aligned} \sum_{r=1}^{n} \big(\ln(r+1) - \ln r\big) &= \big(\cancel{\ln 2} + \cancel{\ln 3} + \cancel{\ln 4} + \cdots + \cancel{\ln n} + \ln(n+1)\big) \\ &\quad - \big(\ln 1 + \cancel{\ln 2} + \cancel{\ln 3} + \cancel{\ln 4} + \cdots + \cancel{\ln n}\big) \\ &= \ln(n+1) - \ln 1 \end{aligned}$$

For the second sum:

$$\sum_{r=1}^{n}\big(\ln(r+2)-\ln(r+3)\big) = \big(\ln 3 + \cancel{\ln 4} + \cancel{\ln 5} + \cdots + \cancel{\ln(n+2)}\big)$$
$$- \big(\cancel{\ln 4} + \cancel{\ln 5} + \cdots + \cancel{\ln(n+2)} + \ln(n+3)\big)$$
$$= \ln 3 - \ln(n+3)$$

Therefore

$$S_n = \big(\ln(n+1) - \ln 1\big) + \big(\ln 3 - \ln(n+3)\big)$$
$$= \ln(n+1) + \ln 3 - \ln(n+3)$$

Combining the logarithms:

$$S_n = \ln\left(\frac{3(n+1)}{n+3}\right)$$

So

$$\sum_{r=1}^{n}\ln\left(1+\frac{2}{r(r+3)}\right) = \ln\left(\frac{3(n+1)}{n+3}\right)$$

(b) Using $\ln\left(x^{28}\right) = 28\ln x$:

$$\sum_{r=25}^{74}\ln\left(\left(1+\frac{2}{r(r+3)}\right)^{28}\right) = 28\sum_{r=25}^{74}\ln\left(1+\frac{2}{r(r+3)}\right)$$

From part (a), if

$$S_n = \sum_{r=1}^{n}\ln\left(1+\frac{2}{r(r+3)}\right)$$

then

$$S_n = \ln\left(\frac{3(n+1)}{n+3}\right)$$

So

$$\sum_{r=25}^{74}\ln\left(1+\frac{2}{r(r+3)}\right) = S_{74} - S_{24}$$
$$= \ln\left(\frac{3(75)}{77}\right) - \ln\left(\frac{3(25)}{27}\right)$$
$$= \ln\left(\frac{225}{77}\right) - \ln\left(\frac{75}{27}\right)$$
$$= \ln\left(\frac{225}{77}\times\frac{27}{75}\right)$$
$$= \ln\left(\frac{81}{77}\right)$$

Hence

$$\sum_{r=25}^{74}\ln\left(\left(1+\frac{2}{r(r+3)}\right)^{28}\right) = 28\ln\left(\frac{81}{77}\right)$$

which is in the form $a\ln\left(\frac{b}{c}\right)$ with $a = 28$, $b = 81$ and $c = 77$.

6. Using partial fractions and a method of differences, prove that for positive integer n,

$$\sum_{r=1}^{n} \frac{6}{r(r+1)(r+2)} = \frac{n(an+b)}{2(n+1)(n+2)}$$

where a and b are constants to be found. **[6]**

Solution

Let

$$S_n = \sum_{r=1}^{n} \frac{6}{r(r+1)(r+2)}$$

Using partial fractions, write

$$\frac{6}{r(r+1)(r+2)} = \frac{A}{r(r+1)} + \frac{B}{(r+1)(r+2)}$$

Multiplying through by $r(r+1)(r+2)$:

$$6 = A(r+2) + Br$$

So

$$6 = (A+B)r + 2A$$

Comparing coefficients gives

$$A + B = 0, \qquad 2A = 6$$

Hence

$$A = 3, \qquad B = -3$$

Therefore

$$\frac{6}{r(r+1)(r+2)} = \frac{3}{r(r+1)} - \frac{3}{(r+1)(r+2)}$$

So

$$S_n = \sum_{r=1}^{n} \left(\frac{3}{r(r+1)} - \frac{3}{(r+1)(r+2)} \right)$$

Now use the method of differences by writing out the first few and last few terms:

$$S_n = \left(\frac{3}{1\times 2} - \cancel{\frac{3}{2\times 3}} \right) + \left(\cancel{\frac{3}{2\times 3}} - \cancel{\frac{3}{3\times 4}} \right) + \left(\cancel{\frac{3}{3\times 4}} - \cancel{\frac{3}{4\times 5}} \right) + \cdots$$
$$+ \left(\cancel{\frac{3}{(n-1)n}} - \cancel{\frac{3}{n(n+1)}} \right) + \left(\cancel{\frac{3}{n(n+1)}} - \frac{3}{(n+1)(n+2)} \right)$$

All the coloured pairs cancel, leaving

$$S_n = \frac{3}{1\times 2} - \frac{3}{(n+1)(n+2)}$$

So

$$S_n = \frac{3}{2} - \frac{3}{(n+1)(n+2)}$$

Put this over the common denominator $2(n+1)(n+2)$:

$$\begin{aligned} S_n &= \frac{3(n+1)(n+2) - 6}{2(n+1)(n+2)} \\ &= \frac{3(n^2+3n+2) - 6}{2(n+1)(n+2)} \\ &= \frac{3n^2+9n}{2(n+1)(n+2)} \\ &= \frac{3n(n+3)}{2(n+1)(n+2)} \end{aligned}$$

We are told this has the form

$$\frac{n(an+b)}{2(n+1)(n+2)}$$

Hence

$$n(an+b) = 3n(n+3) = n(3n+9)$$

Since n is a positive integer, $n \neq 0$, so

$$an + b = 3n + 9$$

Therefore

$$a = 3, \qquad b = 9$$

Hence

$$\sum_{r=1}^{n} \frac{6}{r(r+1)(r+2)} = \frac{3n(n+3)}{2(n+1)(n+2)} = \frac{n(3n+9)}{2(n+1)(n+2)}$$

Alternative Solution:

Write

$$\frac{6}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$$

Using the cover-up rule:

$$A = \frac{6}{(1)(2)} = 3, \qquad B = \frac{6}{(-1)(1)} = -6, \qquad C = \frac{6}{(-2)(-1)} = 3$$

So

$$\frac{6}{r(r+1)(r+2)} = \frac{3}{r} - \frac{6}{r+1} + \frac{3}{r+2}$$

Write out the terms for each value of r:

$$
\begin{aligned}
r = 1: &\quad \frac{3}{1} - \frac{6}{2} + \frac{3}{3} \\
r = 2: &\quad \frac{3}{2} - \frac{6}{3} + \frac{3}{4} \\
r = 3: &\quad \frac{3}{3} - \frac{6}{4} + \frac{3}{5} \\
r = 4: &\quad \frac{3}{4} - \frac{6}{5} + \frac{3}{6} \\
&\quad \vdots \\
r = n-2: &\quad \frac{3}{n-2} - \frac{6}{n-1} + \frac{3}{n} \\
r = n-1: &\quad \frac{3}{n-1} - \frac{6}{n} + \frac{3}{n+1} \\
r = n: &\quad \frac{3}{n} - \frac{6}{n+1} + \frac{3}{n+2}
\end{aligned}
$$

For each k from 3 to n, the terms with denominator k are $\frac{3}{k}$, $-\frac{6}{k}$ and $\frac{3}{k}$, from the rows $r = k-2$, $r = k-1$ and $r = k$. These sum to zero, so they cancel. The terms left are

$$
\begin{aligned}
S_n &= \frac{3}{1} - \frac{6}{2} + \frac{3}{2} + \frac{3}{n+1} - \frac{6}{n+1} + \frac{3}{n+2} \\
&= \frac{3}{2} - \frac{3}{n+1} + \frac{3}{n+2} \\
&= \frac{3}{2} + \frac{-3(n+2) + 3(n+1)}{(n+1)(n+2)} \\
&= \frac{3}{2} - \frac{3}{(n+1)(n+2)}
\end{aligned}
$$

This is the same expression as in the first method, so again

$$
\sum_{r=1}^{n} \frac{6}{r(r+1)(r+2)} = \frac{n(3n+9)}{2(n+1)(n+2)}
$$

with $a = 3$ and $b = 9$.

7. Let $T_n = (n+1)^3 + (n+2)^3 + \cdots + (2n)^3$.

(a) By considering $(r+1)^4 - (r-1)^4$, use the method of differences to prove that

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$ **[5]**

(b) Show that

$$T_n = \frac{1}{4}n^2(an^2 + bn + c)$$

where a, b and c are integers to be determined. **[3]**

(c) State the value of

$$\lim_{n\to\infty} \frac{T_n}{n^4}$$

[1]

Solution

(a) First expand

$$(r+1)^4 - (r-1)^4 = \left(r^4 + 4r^3 + 6r^2 + 4r + 1\right) - \left(r^4 - 4r^3 + 6r^2 - 4r + 1\right) = 8r^3 + 8r$$

Now sum from $r = 1$ to n:

$$\sum_{r=1}^{n}\left((r+1)^4 - (r-1)^4\right) = 8\sum_{r=1}^{n} r^3 + 8\sum_{r=1}^{n} r$$

Using the method of differences on the left-hand side:

$$\begin{aligned}\sum_{r=1}^{n}\left((r+1)^4 - (r-1)^4\right) &= \left(2^4 + 3^4 + 4^4 + \cdots + (n-1)^4 + n^4 + (n+1)^4\right) \\ &\quad - \left(0^4 + 1^4 + 2^4 + 3^4 + 4^4 + \cdots + (n-1)^4\right) \\ &= (n+1)^4 + n^4 - 1\end{aligned}$$

Therefore

$$(n+1)^4 + n^4 - 1 = 8\sum_{r=1}^{n} r^3 + 8\sum_{r=1}^{n} r$$

Using

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

we get

$$\begin{aligned}
8\sum_{r=1}^{n} r^3 &= (n+1)^4 + n^4 - 1 - 8\left(\frac{1}{2}n(n+1)\right) \\
&= (n+1)^4 + n^4 - 1 - 4n(n+1) \\
&= \left(n^4 + 4n^3 + 6n^2 + 4n + 1\right) + n^4 - 1 - 4n^2 - 4n \\
&= 2n^4 + 4n^3 + 2n^2 \\
&= 2n^2(n+1)^2
\end{aligned}$$

So

$$\sum_{r=1}^{n} r^3 = \frac{2n^2(n+1)^2}{8} = \frac{1}{4}n^2(n+1)^2$$

(b)

$$T_n = (n+1)^3 + (n+2)^3 + \cdots + (2n)^3 = \sum_{r=n+1}^{2n} r^3$$

So

$$\begin{aligned}
T_n &= \sum_{r=1}^{2n} r^3 - \sum_{r=1}^{n} r^3 \\
&= \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{4}n^2(n+1)^2
\end{aligned}$$

Now simplify:

$$\begin{aligned}
T_n &= \frac{1}{4}\left(4n^2(2n+1)^2 - n^2(n+1)^2\right) \\
&= \frac{1}{4}\left(4n^2(4n^2+4n+1) - n^2(n^2+2n+1)\right) \\
&= \frac{1}{4}\left(16n^4 + 16n^3 + 4n^2 - n^4 - 2n^3 - n^2\right) \\
&= \frac{1}{4}\left(15n^4 + 14n^3 + 3n^2\right) \\
&= \frac{1}{4}n^2(15n^2 + 14n + 3)
\end{aligned}$$

Therefore

$$T_n = \frac{1}{4}n^2(an^2 + bn + c) \quad \text{with} \quad a = 15,\ b = 14,\ c = 3$$

(c) From part (b):

$$\frac{T_n}{n^4} = \frac{1}{4}\left(15 + \frac{14}{n} + \frac{3}{n^2}\right)$$

As $n \to \infty$, $\frac{14}{n} \to 0$ and $\frac{3}{n^2} \to 0$, so

$$\lim_{n\to\infty} \frac{T_n}{n^4} = \frac{15}{4}$$

8. **(a)** Show that, for $r \geq 2$,

$$\frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}} = \sqrt{r(r+2)}-\sqrt{r(r-2)}$$ **[2]**

(b) Hence use the method of differences to determine

$$\sum_{r=2}^{n} \frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}}$$

giving your answer in simplest form. **[3]**

(c) Hence show that

$$\sum_{r=3}^{7} \frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}} = A\sqrt{2}+B\sqrt{3}+C\sqrt{7}$$

where A, B and C are integers to be determined. **[2]**

Solution

(a) Rationalise the denominator:

$$\begin{aligned}\frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}} &= \frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}} \times \frac{\sqrt{r(r+2)}-\sqrt{r(r-2)}}{\sqrt{r(r+2)}-\sqrt{r(r-2)}} \\ &= \frac{4r\left(\sqrt{r(r+2)}-\sqrt{r(r-2)}\right)}{r(r+2)-r(r-2)} \\ &= \frac{4r\left(\sqrt{r(r+2)}-\sqrt{r(r-2)}\right)}{4r} \\ &= \sqrt{r(r+2)}-\sqrt{r(r-2)}\end{aligned}$$

as required.

(b) Let

$$u_r = \sqrt{r(r+2)}$$

Then from part (a):

$$\frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}} = u_r - u_{r-2}$$

So, if

$$S_n = \sum_{r=2}^{n} \frac{4r}{\sqrt{r(r+2)}+\sqrt{r(r-2)}}$$

then

$$S_n = \sum_{r=2}^{n} (u_r - u_{r-2})$$

Writing out the terms shows the method of differences clearly:

$$\begin{aligned} S_n &= (u_2 + u_3 + u_4 + u_5 + \cdots + u_{n-2} + u_{n-1} + u_n) \\ &\quad - (u_0 + u_1 + u_2 + u_3 + u_4 + u_5 + \cdots + u_{n-2}) \\ &= u_{n-1} + u_n - u_0 - u_1 \end{aligned}$$

Now

$$u_{n-1} = \sqrt{(n-1)(n+1)}, \quad u_n = \sqrt{n(n+2)}, \quad u_0 = \sqrt{0 \times 2} = 0, \quad u_1 = \sqrt{1 \times 3} = \sqrt{3}$$

Hence

$$\sum_{r=2}^{n} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = \sqrt{(n-1)(n+1)} + \sqrt{n(n+2)} - \sqrt{3}$$

(c) Using part (b) with $n = 7$:

$$\begin{aligned} \sum_{r=2}^{7} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} &= \sqrt{6 \times 8} + \sqrt{7 \times 9} - \sqrt{3} \\ &= \sqrt{48} + \sqrt{63} - \sqrt{3} \\ &= 4\sqrt{3} + 3\sqrt{7} - \sqrt{3} \\ &= 3\sqrt{3} + 3\sqrt{7} \end{aligned}$$

The $r = 2$ term is

$$\frac{4(2)}{\sqrt{2(2+2)} + \sqrt{2(2-2)}} = \sqrt{2 \times 4} - \sqrt{2 \times 0} = \sqrt{8} = 2\sqrt{2}$$

from part (a).

Therefore

$$\begin{aligned} \sum_{r=3}^{7} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} &= \sum_{r=2}^{7} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} - 2\sqrt{2} \\ &= 3\sqrt{3} + 3\sqrt{7} - 2\sqrt{2} \end{aligned}$$

So

$$\sum_{r=3}^{7} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = -2\sqrt{2} + 3\sqrt{3} + 3\sqrt{7}$$

Hence

$$A = -2, \qquad B = 3, \qquad C = 3$$

9. Let a be a positive constant.

(a) Use the method of differences to find

$$\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)(r+a+2)}$$

in terms of n and a. **[4]**

(b) Hence find the value of a for which

$$\sum_{r=1}^{\infty} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{24}$$ **[3]**

Solution

(a) Let

$$S_n = \sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)(r+a+2)}$$

To use the method of differences, first note that for any x

$$\begin{aligned}\frac{1}{x(x+1)} - \frac{1}{(x+1)(x+2)} &= \frac{x+2-x}{x(x+1)(x+2)} \\ &= \frac{2}{x(x+1)(x+2)}\end{aligned}$$

So

$$\frac{1}{x(x+1)(x+2)} = \frac{1}{2}\left(\frac{1}{x(x+1)} - \frac{1}{(x+1)(x+2)}\right)$$

Now put $x = r + a$. Then

$$S_n = \frac{1}{2}\sum_{r=1}^{n}\left(\frac{1}{(r+a)(r+a+1)} - \frac{1}{(r+a+1)(r+a+2)}\right)$$

Expanding the sum:

$$\begin{aligned}S_n = \frac{1}{2}\Bigg[&\frac{1}{(a+1)(a+2)} - \cancel{\frac{1}{(a+2)(a+3)}} \\ &+ \cancel{\frac{1}{(a+2)(a+3)}} - \cancel{\frac{1}{(a+3)(a+4)}} \\ &+ \cancel{\frac{1}{(a+3)(a+4)}} - \cdots - \cancel{\frac{1}{(n+a)(n+a+1)}} \\ &+ \cancel{\frac{1}{(n+a)(n+a+1)}} - \frac{1}{(n+a+1)(n+a+2)}\Bigg]\end{aligned}$$

All the middle terms cancel, leaving

$$S_n = \frac{1}{2}\left(\frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)}\right)$$

Hence

$$\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2}\left(\frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)}\right)$$

(b) From part (a):

$$\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2}\left(\frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)}\right)$$

As $n \to \infty$

$$\frac{1}{(n+a+1)(n+a+2)} \to 0$$

So

$$\sum_{r=1}^{\infty} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2(a+1)(a+2)}$$

Given that this equals $\frac{1}{24}$:

$$\frac{1}{2(a+1)(a+2)} = \frac{1}{24}$$

$$2(a+1)(a+2) = 24$$

$$(a+1)(a+2) = 12$$

$$a^2 + 3a + 2 = 12$$

$$a^2 + 3a - 10 = 0$$

$$(a-2)(a+5) = 0$$

So $a = 2$ or $a = -5$. Since a is positive, $a = 2$.

10. **(a)** Use standard results from the formula booklet to show that

$$\sum_{r=1}^{N} r(2r-1)(2r+1) = \frac{1}{2}N(N+1)(2N^2+2N-1) \quad \textbf{[3]}$$

(b) Express $\dfrac{6r+13}{(2r-1)(2r+3)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^{N} \frac{6r+13}{(2r-1)(2r+3)}\left(\frac{1}{2}\right)^{r+1}$$

in terms of N. **[4]**

(c) Deduce the value of

$$\sum_{r=1}^{\infty} \frac{6r+13}{(2r-1)(2r+3)}\left(\frac{1}{2}\right)^{r+1} \quad \textbf{[2]}$$

Solution

(a) Using

$$(2r-1)(2r+1) = 4r^2 - 1$$

we get

$$r(2r-1)(2r+1) = r(4r^2-1) = 4r^3 - r$$

So

$$\sum_{r=1}^{N} r(2r-1)(2r+1) = 4\sum_{r=1}^{N} r^3 - \sum_{r=1}^{N} r$$

From the formula booklet:

$$\sum_{r=1}^{N} r = \frac{1}{2}N(N+1), \qquad \sum_{r=1}^{N} r^3 = \frac{1}{4}N^2(N+1)^2$$

Hence

$$\begin{aligned}\sum_{r=1}^{N} r(2r-1)(2r+1) &= 4\left(\frac{1}{4}N^2(N+1)^2\right) - \frac{1}{2}N(N+1)\\ &= N^2(N+1)^2 - \frac{1}{2}N(N+1)\\ &= \frac{1}{2}N(N+1)\big(2N(N+1)-1\big)\\ &= \frac{1}{2}N(N+1)(2N^2+2N-1)\end{aligned}$$

(b) Let

$$\frac{6r+13}{(2r-1)(2r+3)} = \frac{A}{2r-1} + \frac{B}{2r+3}$$

Then

$$6r + 13 = A(2r + 3) + B(2r - 1)$$

so

$$6r + 13 = (2A + 2B)r + (3A - B)$$

Comparing coefficients:

$$2A + 2B = 6, \qquad 3A - B = 13$$

so

$$A + B = 3, \qquad 3A - B = 13$$

Adding gives $4A = 16$, so $A = 4$, and then $B = -1$.

Therefore

$$\frac{6r + 13}{(2r - 1)(2r + 3)} = \frac{4}{2r - 1} - \frac{1}{2r + 3}$$

Now let

$$S_N = \sum_{r=1}^{N} \frac{6r + 13}{(2r - 1)(2r + 3)} \left(\frac{1}{2}\right)^{r+1}$$

Then

$$\begin{aligned} S_N &= \sum_{r=1}^{N} \left(\frac{4}{2r - 1} - \frac{1}{2r + 3} \right) \frac{1}{2^{r+1}} \\ &= \sum_{r=1}^{N} \left(\frac{4}{(2r - 1)2^{r+1}} - \frac{1}{(2r + 3)2^{r+1}} \right) \\ &= \sum_{r=1}^{N} \left(\frac{4}{(2r - 1)2^{r+1}} - \frac{4}{(2r + 3)2^{r+3}} \right) \end{aligned}$$

Define

$$u_r = \frac{1}{(2r - 1)2^{r+1}}$$

Then

$$u_{r+2} = \frac{1}{(2r + 3)2^{r+3}}$$

so

$$S_N = 4 \sum_{r=1}^{N} (u_r - u_{r+2})$$

Using the method of differences:

$$\begin{aligned} S_N = 4\Big[&(u_1 + u_2 + \cancel{u_3} + \cancel{u_4} + \cdots + \cancel{u_N}) \\ &- (\cancel{u_3} + \cancel{u_4} + \cdots + \cancel{u_N} + u_{N+1} + u_{N+2}) \Big] \\ = 4&(u_1 + u_2 - u_{N+1} - u_{N+2}) \end{aligned}$$

Now

$$u_1 = \frac{1}{1 \times 2^2} = \frac{1}{4}, \qquad u_2 = \frac{1}{3 \times 2^3} = \frac{1}{24}$$

and

$$u_{N+1} = \frac{1}{(2N+1)2^{N+2}}, \qquad u_{N+2} = \frac{1}{(2N+3)2^{N+3}}$$

So

$$\begin{aligned} S_N &= 4\left(\frac{1}{4} + \frac{1}{24} - \frac{1}{(2N+1)2^{N+2}} - \frac{1}{(2N+3)2^{N+3}}\right) \\ &= \frac{7}{6} - \frac{1}{(2N+1)2^N} - \frac{1}{(2N+3)2^{N+1}} \end{aligned}$$

Combining the two final fractions:

$$\begin{aligned} \frac{1}{(2N+1)2^N} + \frac{1}{(2N+3)2^{N+1}} &= \frac{2}{(2N+1)2^{N+1}} + \frac{1}{(2N+3)2^{N+1}} \\ &= \frac{2(2N+3) + (2N+1)}{(2N+1)(2N+3)2^{N+1}} \\ &= \frac{6N+7}{(2N+1)(2N+3)2^{N+1}} \end{aligned}$$

Hence

$$\sum_{r=1}^{N} \frac{6r+13}{(2r-1)(2r+3)}\left(\frac{1}{2}\right)^{r+1} = \frac{7}{6} - \frac{1}{(2N+1)2^N} - \frac{1}{(2N+3)2^{N+1}}$$

or equivalently

$$\sum_{r=1}^{N} \frac{6r+13}{(2r-1)(2r+3)}\left(\frac{1}{2}\right)^{r+1} = \frac{7}{6} - \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}$$

(c) From part (b):

$$S_N = \frac{7}{6} - \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}$$

As $N \to \infty$, the second term tends to 0 because the denominator contains the factor 2^{N+1}.

Therefore

$$\sum_{r=1}^{\infty} \frac{6r+13}{(2r-1)(2r+3)}\left(\frac{1}{2}\right)^{r+1} = \frac{7}{6}$$

11. **(a)** Show that

$$(2n)^5 - (2n-2)^5 \equiv 10(2n-1)^4 + 20(2n-1)^2 + 2$$ **[2]**

(b) Hence, using the method of differences, show that for all positive integer values of n,

$$\sum_{r=1}^{n}(2r-1)^4 = \frac{1}{15}n(2n-1)(2n+1)(an^2+bn+c)$$

where a, b and c are integers to be determined. **[7]**

Solution

(a) Let $m = 2n-1$. Then

$$2n = m+1, \qquad 2n-2 = m-1$$

so

$$\begin{aligned}(2n)^5 - (2n-2)^5 &= (m+1)^5 - (m-1)^5 \\ &= \left(m^5 + 5m^4 + 10m^3 + 10m^2 + 5m + 1\right) \\ &\quad - \left(m^5 - 5m^4 + 10m^3 - 10m^2 + 5m - 1\right) \\ &= 10m^4 + 20m^2 + 2\end{aligned}$$

Substituting back $m = 2n-1$:

$$(2n)^5 - (2n-2)^5 = 10(2n-1)^4 + 20(2n-1)^2 + 2$$

as required.

(b) Let

$$S_n = \sum_{r=1}^{n}(2r-1)^4$$

Using the result from part (a) with n replaced by r:

$$(2r)^5 - (2r-2)^5 = 10(2r-1)^4 + 20(2r-1)^2 + 2$$

Now sum from $r = 1$ to n:

$$\sum_{r=1}^{n}\left((2r)^5 - (2r-2)^5\right) = 10\sum_{r=1}^{n}(2r-1)^4 + 20\sum_{r=1}^{n}(2r-1)^2 + 2n$$

That is

$$\sum_{r=1}^{n}\left((2r)^5 - (2r-2)^5\right) = 10S_n + 20\sum_{r=1}^{n}(2r-1)^2 + 2n$$

For the left-hand side, use the method of differences:

$$\begin{aligned}\sum_{r=1}^{n}\left((2r)^5-(2r-2)^5\right) &= \left(2^5+4^5+6^5+\cdots+\cancel{(2n-2)^5}+(2n)^5\right)\\ &\quad -\left(0^5+2^5+\cancel{4^5}+\cancel{6^5}+\cdots+\cancel{(2n-2)^5}\right)\\ &= (2n)^5\\ &= 32n^5\end{aligned}$$

So

$$32n^5 = 10S_n + 20\sum_{r=1}^{n}(2r-1)^2 + 2n$$

The standard results are

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1), \qquad \sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$

Now find $\sum_{r=1}^{n}(2r-1)^2$:

$$\begin{aligned}\sum_{r=1}^{n}(2r-1)^2 &= \sum_{r=1}^{n}(4r^2-4r+1)\\ &= 4\sum_{r=1}^{n}r^2 - 4\sum_{r=1}^{n}r + \sum_{r=1}^{n}1\\ &= 4\times\frac{1}{6}n(n+1)(2n+1) - 4\times\frac{1}{2}n(n+1)+n\\ &= \frac{2n(n+1)(2n+1)}{3} - 2n(n+1)+n\\ &= \frac{2n(n+1)(2n+1)-6n(n+1)+3n}{3}\\ &= \frac{n(2(n+1)(2n+1)-6(n+1)+3)}{3}\\ &= \frac{n(4n^2-1)}{3}\\ &= \frac{n(2n-1)(2n+1)}{3}\end{aligned}$$

Substitute this into the equation for S_n:

$$32n^5 = 10S_n + 20\left(\frac{n(2n-1)(2n+1)}{3}\right) + 2n$$

Hence

$$\begin{aligned}10S_n &= 32n^5 - \frac{20n(2n-1)(2n+1)}{3} - 2n \\ &= 32n^5 - \frac{20n(4n^2-1)}{3} - 2n \\ &= 32n^5 - \frac{80}{3}n^3 + \frac{20}{3}n - 2n \\ &= 32n^5 - \frac{80}{3}n^3 + \frac{14}{3}n\end{aligned}$$

So

$$\begin{aligned}S_n &= \frac{1}{10}\left(32n^5 - \frac{80}{3}n^3 + \frac{14}{3}n\right) \\ &= \frac{48n^5 - 40n^3 + 7n}{15}\end{aligned}$$

Now factorise:

$$\begin{aligned}48n^5 - 40n^3 + 7n &= n(48n^4 - 40n^2 + 7) \\ &= n(4n^2-1)(12n^2-7) \\ &= n(2n-1)(2n+1)(12n^2-7)\end{aligned}$$

Therefore

$$\sum_{r=1}^{n}(2r-1)^4 = \frac{1}{15}n(2n-1)(2n+1)(12n^2-7)$$

Comparing with

$$\sum_{r=1}^{n}(2r-1)^4 = \frac{1}{15}n(2n-1)(2n+1)(an^2+bn+c)$$

gives

$$a = 12, \qquad b = 0, \qquad c = -7$$

12. Let

$$S_n = \sum_{r=1}^{n} \frac{1}{(3r-2)(3r+4)}$$

where $n \geq 1$.

(a) Use the method of differences to show that

$$S_n = \frac{n(15n+17)}{8(3n+1)(3n+4)} \quad \textbf{[5]}$$

(b) Show that, for any number k greater than $\frac{24}{5}$, if the difference between $\frac{5}{24}$ and S_n is less than $\frac{1}{k}$, then

$$n > \frac{k - 15 + \sqrt{k^2+81}}{18} \quad \textbf{[6]}$$

Solution

(a) Write

$$\frac{1}{(3r-2)(3r+4)} = \frac{A}{3r-2} + \frac{B}{3r+4}$$

Then

$$1 = A(3r+4) + B(3r-2)$$

Comparing coefficients of r and constants gives

$$3A + 3B = 0, \qquad 4A - 2B = 1$$

So

$$A + B = 0 \implies B = -A$$

and hence

$$4A - 2(-A) = 1 \implies 6A = 1 \implies A = \frac{1}{6}, \quad B = -\frac{1}{6}$$

Therefore

$$\frac{1}{(3r-2)(3r+4)} = \frac{1}{6}\left(\frac{1}{3r-2} - \frac{1}{3r+4}\right)$$

So

$$S_n = \frac{1}{6}\sum_{r=1}^{n}\left(\frac{1}{3r-2} - \frac{1}{3r+4}\right)$$

Now expand the sum:

$$S_n = \frac{1}{6}\left[\left(1 - \frac{1}{7}\right) + \left(\frac{1}{4} - \frac{1}{10}\right) + \left(\frac{1}{7} - \frac{1}{13}\right) + \left(\frac{1}{10} - \frac{1}{16}\right) + \cdots \right.$$
$$\left. + \left(\frac{1}{3n-5} - \frac{1}{3n+1}\right) + \left(\frac{1}{3n-2} - \frac{1}{3n+4}\right)\right]$$

Rearranging to show the cancellation pattern:

$$S_n = \frac{1}{6}\left(1 + \frac{1}{4} + \cancel{\frac{1}{7}} + \cancel{\frac{1}{10}} + \cdots + \cancel{\frac{1}{3n-2}}\right.$$
$$\left. - \cancel{\frac{1}{7}} - \cancel{\frac{1}{10}} - \cdots - \cancel{\frac{1}{3n-2}} - \frac{1}{3n+1} - \frac{1}{3n+4}\right)$$

The coloured middle terms cancel, leaving

$$S_n = \frac{1}{6}\left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4}\right)$$

Now simplify:

$$S_n = \frac{1}{6}\left(\frac{5}{4} - \frac{(3n+4)+(3n+1)}{(3n+1)(3n+4)}\right)$$

$$S_n = \frac{1}{6}\left(\frac{5}{4} - \frac{6n+5}{(3n+1)(3n+4)}\right)$$

Putting the terms inside the bracket over a common denominator:

$$\begin{aligned} S_n &= \frac{1}{6}\left(\frac{5(3n+1)(3n+4) - 4(6n+5)}{4(3n+1)(3n+4)}\right) \\ &= \frac{1}{6}\left(\frac{5(9n^2+15n+4) - 24n - 20}{4(3n+1)(3n+4)}\right) \\ &= \frac{1}{6}\left(\frac{45n^2+75n+20-24n-20}{4(3n+1)(3n+4)}\right) \\ &= \frac{1}{6}\left(\frac{45n^2+51n}{4(3n+1)(3n+4)}\right) \\ &= \frac{1}{6}\left(\frac{3n(15n+17)}{4(3n+1)(3n+4)}\right) \end{aligned}$$

Hence

$$S_n = \frac{n(15n+17)}{8(3n+1)(3n+4)}$$

(b) From part (a):

$$S_n = \frac{1}{6}\left(\frac{5}{4} - \frac{6n+5}{(3n+1)(3n+4)}\right) = \frac{5}{24} - \frac{6n+5}{6(3n+1)(3n+4)}$$

So

$$\frac{5}{24} - S_n = \frac{6n+5}{6(3n+1)(3n+4)}$$

If this difference is less than $\frac{1}{k}$, then

$$\frac{6n+5}{6(3n+1)(3n+4)} < \frac{1}{k}$$

Since $k > \frac{24}{5} > 0$, all quantities are positive, so we may cross-multiply:

$$k(6n+5) < 6(3n+1)(3n+4)$$

Expanding:

$$6kn + 5k < 6(9n^2 + 15n + 4) = 54n^2 + 90n + 24$$

Hence

$$54n^2 + (90 - 6k)n + (24 - 5k) > 0$$

Now solve the corresponding quadratic equation

$$54n^2 + (90 - 6k)n + (24 - 5k) = 0$$

Using the quadratic formula:

$$n = \frac{-(90-6k) \pm \sqrt{(90-6k)^2 - 4(54)(24-5k)}}{108}$$

Simplify the discriminant:

$$\begin{aligned}(90-6k)^2 - 4(54)(24-5k) &= 36(15-k)^2 - 216(24-5k) \\ &= 36\big((15-k)^2 - 6(24-5k)\big) \\ &= 36\big(k^2 - 30k + 225 - 144 + 30k\big) \\ &= 36(k^2+81)\end{aligned}$$

Therefore

$$\begin{aligned}n &= \frac{6k - 90 \pm 6\sqrt{k^2+81}}{108} \\ &= \frac{k - 15 \pm \sqrt{k^2+81}}{18}\end{aligned}$$

Since the coefficient of n^2 is positive, the inequality

$$54n^2 + (90 - 6k)n + (24 - 5k) > 0$$

holds for values of n outside the two roots.

Also

$$k - 15 - \sqrt{k^2+81} < k - 15 - k = -15 < 0$$

So the smaller root is negative. As $n \geq 1$, it follows that

$$n > \frac{k - 15 + \sqrt{k^2+81}}{18}$$

as required.